its parent category. This approach allows information to flow from parent to child categories, effectively leveraging the hierarchical structure of the data and enhancing prediction accuracy.

5.3 *Evaluation Metrics*

Following Faust and Wright [2013] and Aparicio and Bertolotto [2020], we report our results using three evaluation metrics:

1. **Root Mean Squared Error (RMSE) -** The RMSE is calculated as:

$$RMSE = \sqrt{\frac{1}{T}\sum_{t=1}^{T}(x_t - \hat{x}_t)^2}, \tag{5.1}$$

   where x_t represents the actual monthly change rate for month t, and $\hat{x}_t$ denotes the corresponding predicted value.

2. **Pearson Correlation Coefficient -** The Pearson correlation coefficient ϕ is defined as:

$$\phi = \frac{COV(X_T, \hat{X}_T)}{\sigma_X \times \sigma_{\hat{X}}}, \tag{5.2}$$

   where $COV(X_T, \hat{X}_T)$ is the covariance between the actual values and predictions, and σ_{X_T} and $\sigma_{\hat{X}_T}$ are the standard deviations of the actual values and the predictions, respectively.

3. **Distance Correlation Coefficient -** Unlike the Pearson correlation, which only measures linear relationships, the distance correlation coefficient can detect both linear and nonlinear associations [Székely et al., 2007, Zhou, 2012]. The distance correlation coefficient r_d is given by:

$$r_d = \frac{\mathrm{dCov}(X_T, \hat{X}_T)}{\sqrt{\mathrm{dVar}(X_T) \times \mathrm{dVar}(\hat{X}_T)}} \tag{5.3}$$

   where $\mathrm{dCov}(X_T, \hat{X}_T)$ is the distance covariance between the actual values and the predictions, and $\mathrm{dVar}(X_T)$ and $\mathrm{dVar}(\hat{X}_T)$ are the distance variances of the actual values and the predictions, respectively.

5.4 *Results*

The BiHRNN model stands out for its ability to leverage information flow both from higher levels to lower levels and from lower levels to higher levels within the hierarchy. The model leverages the inherent hierarchy of the CPI, enhancing predictions at both granular and broader, more significant levels, such as the CPI Headline. Therefore, we will provide the headline results separately, along with the aggregated results across all categories.

The results are relative to the $AR(1)$ model and normalized according to: $\frac{RMSE_{Model}}{RMSE_{AR(1)}}$.

5.4.1 *US CPI Results*

Table 5.1: Average Results on Disaggregated CPI Components - US

Model	Avg. RMSE	Pearson Corr.	Dist. Corr.	Headline RMSE	Headline Pearson Corr.	Headline Dist. Corr.
I-GRU	1.215	0.138	0.338	1.015	0.347	0.350
AR_1	1.000	0.176	0.513	1.000	0.327	0.459
AR_2	1.267	0.105	0.467	1.312	0.327	0.565
AR_3	1.487	0.082	0.437	1.560	0.349	0.510
AR_4	1.902	0.052	0.411	1.749	0.308	0.427
FC_p_12	1.229	-0.014	0.355	1.592	0.027	0.251
RF_p_12	1.143	0.112	0.377	1.210	0.368	0.377
RW_p_4	1.189	-0.013	0.353	1.420	-0.050	0.310
SVR_p_12	1.115	0.067	0.363	1.280	0.473	0.529
XGB_p_12	1.228	0.087	0.369	1.312	0.392	0.423
HRNN	1.028	0.158	0.346	1.015	0.347	0.350
BiHRNN	0.966	0.230	0.378	1.052	0.225	0.290

5.4.2 *Canada CPI Results*

Table 5.2: Average Results on Disaggregated CPI Components - Canada

Model	Avg. RMSE	Pearson Corr.	Dist. Corr.	Headline RMSE	Headline Pearson Corr.	Headline Dist. Corr.
I-GRU	0.892	0.351	0.476	1.261	0.329	0.516
AR_1	1.000	0.128	0.633	1.000	0.628	0.693
AR_2	1.188	-0.008	0.592	1.155	0.241	0.409
AR_3	1.305	-0.004	0.561	0.970	0.646	0.671
AR_4	1.684	-0.004	0.522	1.424	0.356	0.416
FC_p_12	0.907	0.237	0.465	2.051	0.152	0.352
RF_p_12	0.861	0.318	0.509	1.635	0.542	0.540
RW_p_4	0.901	0.138	0.414	1.261	0.495	0.658
SVR_p_12	0.852	0.308	0.517	1.721	0.449	0.580
XGB_p_12	0.936	0.271	0.502	1.635	0.411	0.527
HRNN	0.824	0.358	0.490	1.261	0.329	0.516
BiHRNN	0.795	0.386	0.511	1.170	0.321	0.497

5.4.3 Norway CPI Results

Table 5.3: Average Results on Disaggregated CPI Components - Norway

Model	Avg. RMSE	Pearson Corr.	Dist. Corr.	Headline RMSE	Headline Pearson Corr.	Headline Dist. Corr.
I-GRU	0.866	0.355	0.5063	0.724	0.208	0.413
AR_1	1.000	0.053	0.653	1.000	-0.583	0.546
AR_2	1.251	-0.003	0.617	1.132	-0.743	0.695
AR_3	1.378	0.009	0.568	1.189	-0.797	0.777
AR_4	1.727	0.011	0.535	1.296	-0.536	0.574
FC_p_12	0.974	0.226	0.454	0.924	0.242	0.422
RF_p_12	0.849	0.349	0.539	0.672	0.613	0.721
RW_p_4	0.973	0.187	0.405	0.788	0.165	0.337
SVR_p_12	0.851	0.376	0.550	0.848	0.251	0.362
XGB_p_12	0.904	0.303	0.530	0.669	0.644	0.725
HRNN	0.832	0.3677	0.5365	0.724	0.208	0.413
BiHRNN	0.767	0.478	0.567	0.655	0.390	0.477

The results in Tables 5.2, 5.3, and 5.1 show that the BiHRNN consistently outperforms other models in terms of predictive accuracy and stability. With some of the lowest RMSE values across datasets, this model demonstrates its ability to reliably minimize error between various components of the CPI. In comparison, simpler models like AR(1) and AR(4) often exhibit higher RMSE and greater variability, indicating that the BiHRNN model offers a stronger, more stable fit for this complex data.

Correlation metrics reinforce this model's capacity to understand the underlying relationships in the data. The BiHRNN achieves high Pearson and Distance correlations indicating a strong alignment between model predictions and actual outcomes. Although a few models, like RF(12), show competitive correlations, their higher RMSE values demonstrate an inability to consistently maintain accuracy across metrics.

The BiHRNN model demonstrates top-tier performance in headline predictions, excelling in both RMSE and correlation metrics. However, our findings indicate that the Headline data alone is sufficient for accurate headline predictions and yields the best results. Attempts to incorporate additional regularization terms do not enhance prediction performance. Consequently, we recommend that future work on headline predictions focus exclusively on using the Headline data.

This balance across both disaggregated components and headline metrics highlights the model's robustness and adaptability, making it a preferable choice for forecasting CPI trends. Overall, the BiHRNN stands out as the most effective model, combining low error rates, strong fit, and high correlation, all of which contribute to a more accurate and reliable CPI prediction framework across categories.

Figure 5.1 below showcases examples of several disaggregated indexes from different hierarchy levels and sectors. The solid black line shows the actual CPI values, while the dashed lines depict predictions from the top-performing models—all variations of RNN